



Graphical Solution

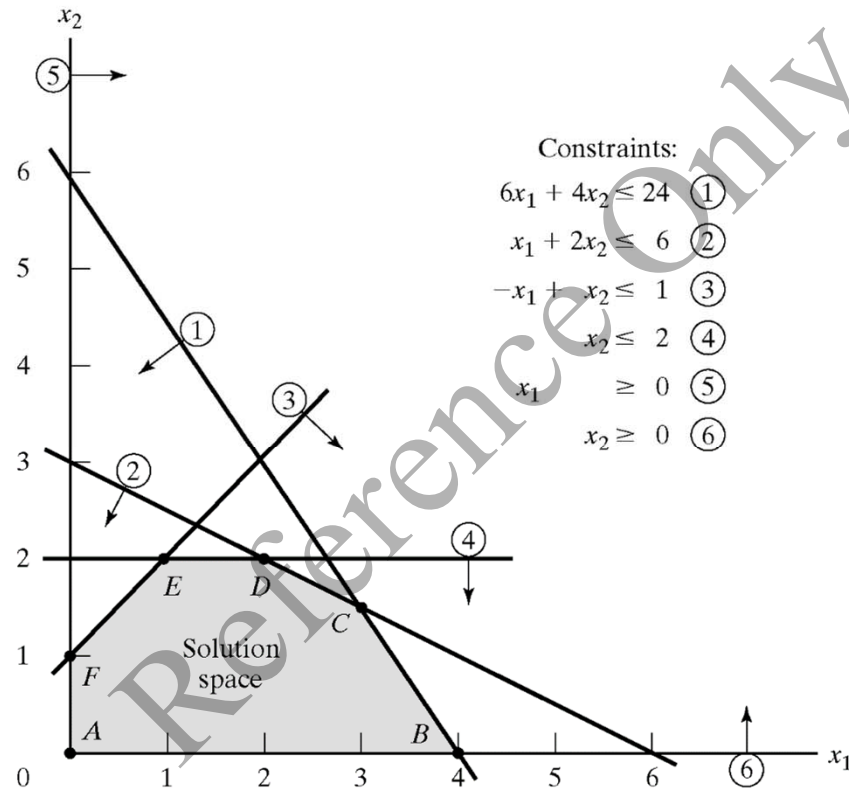


Figure 2.1
Feasible space of the Reddy Mikks model.



Graphical Solution

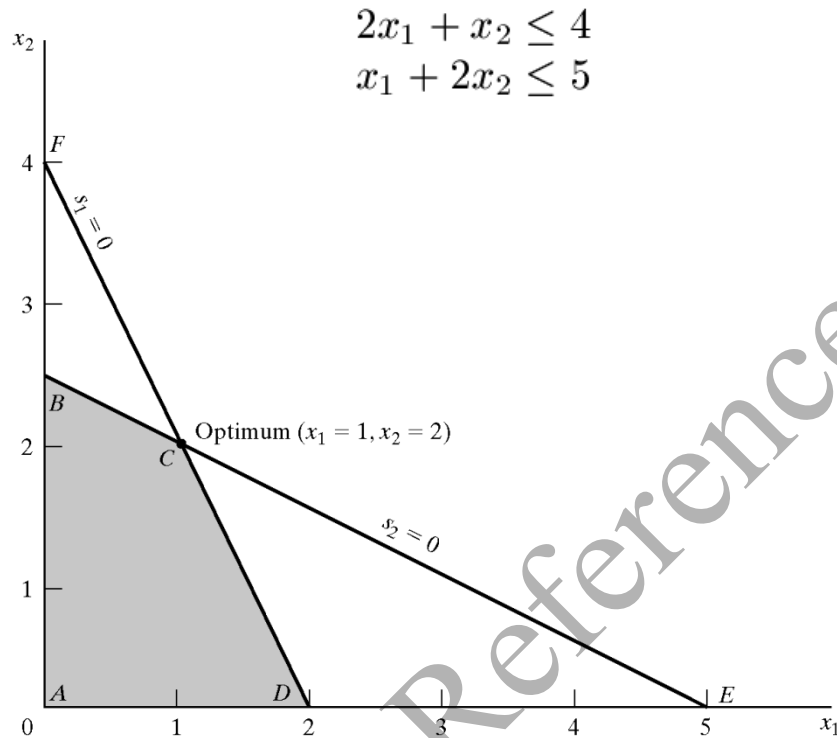


Figure 3.2
LP solution space of Example 3.2-1.

- When $x_1=x_2=0$, corresponds to point A.
- At point A, $s_1=4$, $s_2=5$.



Simplex Algorithm ver 2

1. Put the LP in standard form. * (b vector is non-negative)
2. Make the origin your starting BFS and make the corresponding changes to the LP in standard form.
3. Modify the objective function accordingly.
4. Test for Optimality
5. If optimal, STOP, else go to step 5. * (there are other branches for this step)
6. Determine the Entering Variable.
7. Determine the Leaving Variable.
8. Determine the new BFS.
9. Go to step 3.



Big-M

$$\begin{aligned}
 &\text{Min } 4x_1 + x_2 \\
 &\text{subject to} \\
 &3x_1 + x_2 = 3 \\
 &4x_1 + 3x_2 \geq 6 \\
 &x_1 + 2x_2 \leq 4 \\
 &x_1, x_2 \geq 0
 \end{aligned}$$

	x1	x2	s2	R1	R2	s3	Solution
z	-4	-1	0	-M	-M	0	0
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
s3	1	2	0	0	0	1	4

Express objective function row in terms of the nonbasic variables

$$Mr_1 + Mr_2 + r_0 = r_0$$



Big-M

$$\begin{aligned} \text{Min } & 4x_1 + x_2 \\ \text{subject to } & \\ & 3x_1 + x_2 = 3 \\ & 4x_1 + 3x_2 \geq 6 \\ & x_1 + 2x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$



	x1	x2	s2	R1	R2	s3	Solution
z	7M-4	4M-1	-M	0	0	0	9M
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
s3	1	2	0	0	0	1	4



What is the pivot element?



Big-M

	x1	x2	s2	R1	R2	s3	Solution
z	7M-4	4M-1	-M	0	0	0	9M
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
s3	1	2	0	0	0	1	4

$$\begin{aligned} \frac{1}{3} r_1 &= r_1 \\ -4r_1 + r_2 &= r_2 \\ -r_1 + r_3 &= r_3 \\ -(7M-4)r_1 + r_0 &= r_0 \end{aligned}$$



	x1	x2	s2	R1	R2	s3	Solution
z	0	$5/3M + 1/3$	-M	$-7/3M + 4/3$	0	0	2M+4
x1	1	1/3	0	1/3	0	0	1
R2	0	5/3	-1	-4/3	1	0	2
s3	0	5/3	0	-1/3	0	1	3



What is the pivot element?

L Lorenzo, **UP** IE/OR



Big-M

	x1	x2	s2	R1	R2	s3	Solution
z	0	$5/3M + 1/3$	$-M - 7/3M + 4/3$	0	0	0	$2M + 4$
x1	1	$1/3$	0	$1/3$	0	0	1
R2	0	$5/3$	-1	$-4/3$	1	0	2
s3	0	$5/3$	0	$-1/3$	0	1	3



	x1	x2	s2	R1	R2	s3	Solution
z	0	0	$1/5$	$-M + 8/5$	$-M - 1/5$	0	$18/5$
x1	1	0	$1/5$	$3/5$	$-1/5$	0	$3/5$
x2	0	1	$-3/5$	$-4/5$	$3/5$	0	$6/5$
s3	0	0	1	1	-1	1	1



L Lorenzo, UP IE/OR



Big-M



	x1	x2	s2	R1	R2	s3	Solution
z	0	0	1/5	-M+8/5	-M-1/5	0	18/5
x1	1	0	1/5	3/5	-1/5	0	3/5
x2	0	1	-3/5	-4/5	3/5	0	6/5
s3	0	0	1	1	-1	1	1



	x1	x2	s2	R1	R2	s3	Solution
z	0	0	0	-M+7/5	-M	-1/5	17/5
x1	1	0	0	2/5	0	-1/5	2/5
x2	0	1	0	-1/5	0	3/5	9/5
s2	0	0	1	1	-1	1	1



Simplex Algorithm ver 3

1. Put the LP in standard form. *(b vector is non-negative)
2. Setup the Phase 1 LP and solve using ver 2.
3. If Phase 1 is optimal and all artificial variables are non-basic, proceed to Phase 2.
4. Setup the Phase 2 LP and solve using ver 2.



Two-Phase Method

$$\text{Min } 4x_1 + x_2$$

subject to

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

$$\text{Min } R_1 + R_2$$

subject to

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

	x1	x2	s2	R1	R2	s3	Solution
z	0	0	0	-1	-1	0	0
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
s3	1	2	0	0	0	1	4

Express objective function row in terms of the nonbasic variables

$$r_1 + r_2 + r_0 = r_0$$



Two-Phase Method (Phase I)

$$\text{Min } R_1 + R_2$$

subject to

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

	x1	x2	s2	R1	R2	s3	Solution
z	0	0	0	-1	-1	0	0
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
s3	1	2	0	0	0	1	4

	x1	x2	s2	R1	R2	s3	Solution
z	7	4	-1	0	0	0	9
R1	3	1	0	1	0	0	3
R2	4	3	-1	0	1	0	6
s3	1	2	0	0	0	1	4

What is the pivot element?

L Lorenzo, **UP** IE/OR



Two-Phase Method (Phase II)

	x1	x2	s2	R1	R2	s3	Solution
z	0	0	0	-1	-1	0	0
X1*	1	0	1/5	3/5	-1/5	0	3/5
X2*	0	1	-3/5	-4/5	3/5	0	6/5
S3*	0	0	1	1	-1	1	1

Min $4x_1 + x_2$
 subject to
 $3x_1 + x_2 = 3$
 $4x_1 + 3x_2 \geq 6$
 $x_1 + 2x_2 \leq 4$
 $x_1, x_2 \geq 0$

	x1	x2	s2	s3	Solution
z	-4	-1	0	0	0
X1	1	0	1/5	0	3/5
X2	0	1	-3/5	0	6/5
s3	0	0	1	1	1

Express objective function row in terms of the nonbasic variables

What is the required elementary row operation?



Two-Phase Method (Phase II)

Min $4x_1 + x_2$
 subject to
 $3x_1 + x_2 = 3$
 $4x_1 + 3x_2 \geq 6$
 $x_1 + 2x_2 \leq 4$
 $x_1, x_2 \geq 0$

	x1	x2	s2	s3	Solution
z	-4	-1	0	0	0
x1	1	0	1/5	0	3/5
x2	0	1	-3/5	0	6/5
s3	0	0	1	1	1

	x1	x2	s2	s3	Solution
z	0	0	1/5	0	18/5
x1	1	0	1/5	0	3/5
x2	0	1	-3/5	0	6/5
s3	0	0	1	1	1



Two-Phase Method (Phase II)

Min $4x_1 + x_2$
 subject to
 $3x_1 + x_2 = 3$
 $4x_1 + 3x_2 \geq 6$
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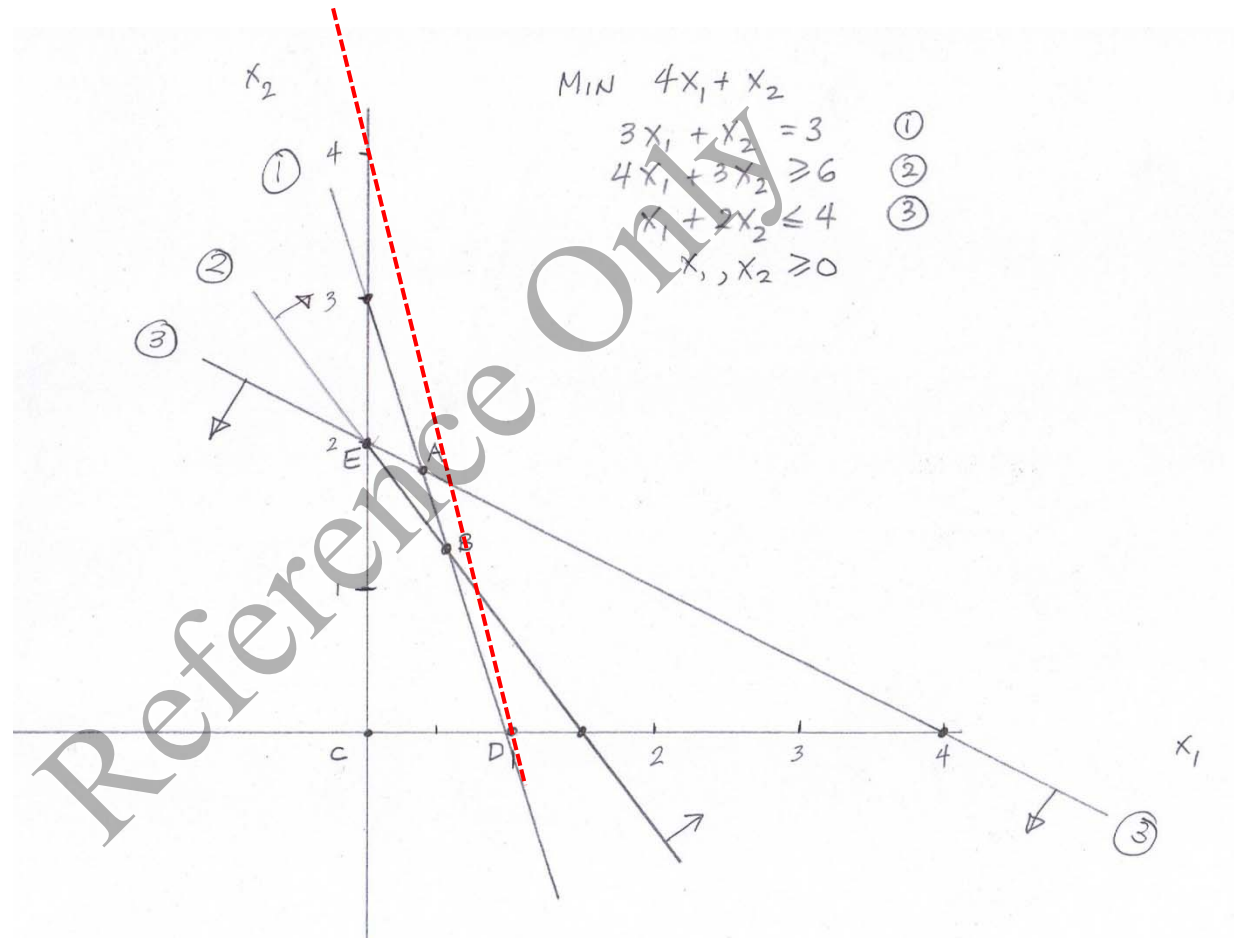
	x1	x2	s2	s3	Solution
z	0	0	1/5	0	18/5
x1	1	0	1/5	0	3/5
x2	0	1	-3/5	0	6/5
s3	0	0	1	1	1

	x1	x2	s2	s3	Solution
z	0	0	0	-1/5	17/5
x1	1	0	0	-1/5	2/5
x2	0	1	0	3/5	9/5
s2	0	0	1	1	1



Two-Phase Method

$$\begin{aligned} \text{Min } & 4x_1 + x_2 \\ \text{subject to } & 3x_1 + x_2 = 3 \\ & 4x_1 + 3x_2 \geq 6 \\ & x_1 + 2x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$





Artificial Variables at Optimality

- To have an optimal solution, all artificial variables must be nonbasic.
- If an artificial variable is basic and positive, the LP is infeasible.
- If an artificial variable is basic and zero, let it leave the basic solution.

Reference Only